Solutions to the exercises in *Categorical Logic and Type Theory* by Bart Jacobs

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Chapter 0

Prospectus

0.2 The logic and type theory of sets

0.2.1. Define a left adjoint $F : \mathrm{Fam}(\mathbf{Sets}) \to \mathbf{Pred}$ to the inclusion functor

$$\begin{tikzcd}
\mathbf{Pred} \arrow[rr, hook] \arrow[dr] & & \mathrm{Fam}(\mathbf{Sets}) \arrow[ll, bend right, "F"'] \arrow[dl] \\
& \mathbf{Sets} &
\end{tikzcd}$$

such that: (1) F makes the triangle commute (so it does not change the index set), and (2) F commutes with substitution.

Solution. Recall that the inclusion functor $G : \mathbf{Pred} \hookrightarrow \mathrm{Fam}(\mathbf{Sets})$ sends a predicate (I, X) (where $X \subseteq I$) to the family $G(I, X) = (X_i)_{i \in I}$, where

$$X_i = \begin{cases} \{*\} & \text{if } i \in X \\ \emptyset & \text{otherwise.} \end{cases}$$

Define the functor $F : \mathrm{Fam}(\mathbf{Sets}) \to \mathbf{Pred}$ on objects by

$$F\left((X_i)_{i \in I}\right) = \left(I, \left\{i \in I : X_i \neq \emptyset\right\}\right)$$

and on morphisms by

$$F\left((X_i)_{i \in I} \xrightarrow{(u,f)} (Y_j)_{j \in J}\right) = u.$$

Note that F is well-defined on morphisms. Indeed, suppose $(X_i)_{i \in I}$ and $(Y_j)_{j \in J}$ are families of sets and $(u, f) : (X_i)_{i \in I} \to (Y_j)_{j \in J}$ is a morphism of families. If $i \in I$ and $X_i \neq \emptyset$, then $\emptyset \neq f(X_i) \subseteq Y_{u(i)}$, so $Y_{u(i)} \neq \emptyset$, and hence

u is a morphism from $F\left((X_i)_{i\in I}\right)$ to $F\left((Y_j)_{j\in J}\right)$ in **Pred**. Moreover, F does not change the index set by definition.

Next, we claim that $F \dashv G$ by demonstrating the adjointness criteria in [2, Definition 3.1.4], namely that there is a "unit" natural transformation

$$\eta : \mathrm{id}_{\mathrm{Fam}(\mathbf{Sets})} \longrightarrow G \circ F.$$

such that for each I-indexed family of sets $(X_i)_{i\in I}$, each object (J, Y) in **Pred**, and each morphism $(v, g) : (X_i)_{i\in I} \to G(J, Y)$ in Fam(**Sets**), there exists a unique morphism $v' : F\left((X_i)_{i\in I}\right) \to (J, Y)$ in **Pred** such that $G(v') \circ \eta_{(X_i)_{i\in I}} = (v, g)$, as in the following commutative diagram.

$$\begin{tikzcd}
\textbf{Pred} & \mathrm{Fam}(\textbf{Sets}) \\
F\left((X_i)_{i\in I}\right) \arrow[d, dashed, "v'"] & (X_i)_{i\in I} \arrow[r, "\eta_{(X_i)_{i\in I}}"] \arrow[dr, "(v,g)"'] & G\left(F\left((X_i)_{i\in I}\right)\right) \arrow[d, dashed, "G(v')"] \\
(J, Y) & & G(J, Y)
\end{tikzcd}$$

To this end, for an object $(X_i)_{i\in I}$ define the component

$$\eta_{(X_i)_{i\in I}} : (X_i)_{i\in I} \longrightarrow G\left(F\left((X_i)_{i\in I}\right)\right) = \left(\text{if } X_i \neq \emptyset \text{ then } \{*\} \text{ else } \emptyset\right)_{i\in I}$$

as

$$\eta_{(X_i)_{i\in I}} = \left(\mathrm{id}_I, \left(f_i : X_i \longrightarrow \text{if } X_i \neq \emptyset \text{ then } \{*\} \text{ else } \emptyset\right)_{i\in I}\right),$$

where

$$f_i = \begin{cases} x \mapsto * & \text{if } X_i \neq \emptyset \\ \mathrm{id}_\emptyset & \text{otherwise.} \end{cases}$$

We omit the rest of the verification that η is the unit of the adjunction $F \dashv G$.

Finally, we claim that F commutes with substitution. Indeed, if $u : I \to J$ is a morphism in **Sets** and $(Y_i)_{j\in J}$ is a J-indexed family of sets, then

$$\begin{aligned}
u^*\left(F\left((Y_j)_{j\in J}\right)\right) &= u^*\left(J, \left\{j \in J : Y_j \neq \emptyset\right\}\right) \\
&= \left(I, \left\{i \in I : Y_{u(i)} \neq \emptyset\right\}\right) \\
&= F\left(\left(Y_{u(i)}\right)_{i\in I}\right) \\
&= F\left(u^*\left((Y_j)_{j\in J}\right)\right).
\end{aligned}$$

□

0.2.2. Define for a subset $X \subseteq I$ the relation $\mathrm{nEq}(X) \subseteq I \times I$ by

$$\mathrm{nEq}(X) = \Big\{(i, i') : i \neq i' \text{ or } i \in X\Big\}$$

and show that the assignment $X \mapsto \mathrm{nEq}(X)$ is right adjoint to contraction $\delta^* : P(I \times I) \to P(I)$. Notice that $\mathrm{nEq}(X)$ at the bottom element $X = \emptyset$ is inequality on I.

Solution. Fix a set I and consider the functors

$$P(I) \underset{\mathrm{nEq}}{\overset{\delta^*}{\leftrightarrows}} P(I \times I).$$

To show that $\delta^* \dashv \mathrm{nEq}$, we fix $X \subseteq I$ and $Y \subseteq I \times I$ and claim that $\delta^*(Y) \subseteq X$ if and only if $Y \subseteq \mathrm{nEq}(X)$. Suppose $\delta^*(Y) \subseteq X$ and $(i, j) \in Y$. If $i \neq j$, then $(i, j) \in \mathrm{nEq}(X)$, and if $i = j$, then $i \in \delta^*(Y) \subseteq X$, and hence $(i, j) \in \mathrm{nEq}(X)$. Thus, $Y \subseteq \mathrm{nEq}(X)$. Conversely, suppose $Y \subseteq \mathrm{nEq}(X)$ and $i \in \delta^*(Y)$. Then $(i, i) \in Y \subseteq \mathrm{nEq}(X)$, so $i \in X$, whence $\delta^*(Y) \subseteq X$ and $(i, j) \in Y$. $\square$

0.2.3. Show that the equality functor $\mathrm{Eq} : \mathbf{Sets} \to \mathbf{Rel}$ also has a right adjoint.

Solution. Define the functor

$$F : \mathbf{Rel} \longrightarrow \mathbf{Sets}$$

on objects by

$$F(I, R) = \{i \in I : (i, i) \in R\}$$

and on morphisms by

$$F\Big((I, R) \xrightarrow{u} (J, S)\Big) = u|_{F(I,R)}.$$

Note that F is well-defined on morphisms since if $u : (I, R) \to (J, S)$ is a morphism in $\mathbf{Rel}$ and $i \in F(I, R)$, then $(i, i) \in R$, so $(u(i), u(i)) \in S$, and hence $u(i) \in F(J, S)$. We claim that $\mathrm{Eq} \dashv F$. Fix an object I in $\mathbf{Sets}$ and an object (J, S) in $\mathbf{Rel}$. Define functions

$$f : \mathbf{Rel}(\mathrm{Eq}(I), (J, S)) \longrightarrow \mathbf{Sets}(I, F(J, S))$$
$$g : \mathbf{Sets}(I, F(J, S)) \longrightarrow \mathbf{Rel}(\mathrm{Eq}(I), (J, S))$$

by

$$f\left(\mathrm{Eq}(I) \xrightarrow{u} (J,S)\right) = u \qquad\qquad g\left(I \xrightarrow{u} F(J,S)\right) = u.$$

Note that f is well-defined since if $u : \mathrm{Eq}(I) \to (J,S)$ is a morphism in **Rel** and $i \in I$, then $(i,i) \in \mathrm{Eq}(I)$, so $(u(i), u(i)) \in S$, and hence $u(i) \in F(J,S)$. Similarly, note that g is well-defined since if $u : I \to F(J,S)$ is a morphism in **Sets** and $i \in I$, then $u(i) \in F(J,S)$, so $(u(i), u(i)) \in S$. Thus, f and g are mutual inverses, so we have

$$\mathbf{Rel}(\mathrm{Eq}(I), (J,S)) \cong \mathbf{Sets}(I, F(J,S)).$$

We omit the proof of naturality of this bijection and conclude that $\mathrm{Eq} \dashv F$. $\square$

0.2.4. Check that the operation $(I,X) \mapsto \{I \mid X\}$ yields a functor $\mathrm{Fam}(\mathbf{Sets}) \to \mathbf{Sets}$, and show that it is right adjoint to the terminal object functor $\mathbf{Sets} \to \mathrm{Fam}(\mathbf{Sets})$, mapping a set J to the family of singletons $(\{*\})_{j\in J}$. Describe the unit and counit of the adjunction explicitly.

Solution. Define the functor $R : \mathrm{Fam}(\mathbf{Sets}) \to \mathbf{Sets}$ on objects by

$$R(I,X) = \{I \mid X\} = \sum_{i\in I} X_i = \Big\{(i,x) : i \in I \text{ and } x \in X_i\Big\}$$

and on morphisms by

$$R\left((I,X) \xrightarrow{(u,f)} (J,Y)\right) = \Big((i,x) \longmapsto (u(i), f_i(x))\Big).$$

We claim that $L \dashv R$, where $L : \mathbf{Sets} \to \mathrm{Fam}(\mathbf{Sets})$ is the "terminal object functor," defined on objects by $L(J) = (\{*\})_{j\in J}$. It suffices to prove that there is an isomorphism

$$\mathrm{Fam}(\mathbf{Sets})\left((\{*\})_{j\in J}, (I,X)\right) \cong \mathbf{Sets}\left(J, \{I \mid X\}\right),$$

which is natural in J in **Sets** and (I,X) in $\mathrm{Fam}(\mathbf{Sets})$. Indeed, define

$$\varphi : \mathrm{Fam}(\mathbf{Sets})\left((\{*\})_{j\in J}, (I,X)\right) \longrightarrow \mathbf{Sets}\left(J, \{I \mid X\}\right)$$

by

$$\varphi\left(\left(\{*\}\right)_{j\in J} \xrightarrow{(u,f)} (I,X)\right) = \Big(j \longmapsto (u(j), f_j(*))\Big).$$

One can verify that φ is a bijection, with

$$\varphi^{-1}\left(J \xrightarrow{g} \{I \mid X\}\right) = \left(j \longmapsto \mathrm{fst}(g(j)), \Big(* \longmapsto \mathrm{snd}(g(j))\Big)_{j\in J}\right),$$

and that φ is natural in J and (I,X). Thus, it yields an adjunction $L \dashv R$. The unit of this adjunction is then given by

$$\eta_J = \varphi\left(\mathrm{id}_{(\{*\})_{j\in J}}\right) : J \longrightarrow \sum_{j\in J}\{*\}, \qquad \eta_J(j) = (j,*),$$

and the counit is given by

$$\begin{aligned}
&\varepsilon_{(I,X)} : (\{*\})_{(i,x)\in\{I|X\}} \longrightarrow (I,X)\\
&\varepsilon_{(I,X)} = \varphi^{-1}\left(\mathrm{id}_{\{I|X\}}\right) = \left((i,x) \longmapsto i, \Big(* \longmapsto x\Big)_{(i,x)\in\{I|X\}}\right). \qquad \square
\end{aligned}$$

Chapter 1

Introduction to fibred category theory

1.1 Fibrations

1.1.1.

(i) Prove Proposition 1.1.4.[a]

(ii) Suppose f is Cartesian and g and h are above the same map. Show that $f \circ g = f \circ h$ implies $g = h$.

[a] [6, Proposition 1.1.4] asserts that Cartesian liftings are unique up-to-isomorphism (in a slice): if f and f' with $\operatorname{cod} f = \operatorname{cod} f'$ are both Cartesian over the same map, then there is a unique vertical isomorphism $\varphi : X \xrightarrow{\cong} X'$ with $f' \circ \varphi = f$.

Remark 1.1.1.1. We will prove this in the more general context of displayed categories (cf. [1]). The conclusion of Exercise 1.1.1(i) (i.e., [6, Proposition 1.1.4]) appears in the literature in, e.g., [3, Lemma 8.1.4(2)], [4, Remark 3.3], [5, Lemma 2.3(ii)], and [7, Proposition 9.1.4(4)].

Solution, (i). Suppose $\mathbb{E}$ is a displayed category over $\mathbb{B}$, $u : I \to J$ is a morphism in $\mathbb{B}$, and $f : X \to Y$ and $f' : X' \to Y$ are Cartesian morphisms in $\mathbb{E}$ over u. Since f' is Cartesian and f is a morphism into Y over $u = u \circ \mathrm{id}_I$, there exists a unique morphism $\varphi : X \to X'$ in $\mathbb{E}$ over I such that $f = f' \circ \varphi$. This is summarized by the following diagram.

$$
\begin{tikzcd}
X \arrow[d] \arrow[rr, bend left, "f"] & & \\
& X' \arrow[r, "f'"] \arrow[d] \arrow[dr, bend left, "u"] & Y \arrow[d] \\
I \arrow[r, equal] & I \arrow[r, "u"] & J
\end{tikzcd}
\quad \Longrightarrow \quad
\begin{tikzcd}
X \arrow[d] \arrow[rr, bend left, "f"] \arrow[dr, dashed, "\exists!\,\varphi"] & & \\
& X' \arrow[r, "f'"] \arrow[d] & Y \arrow[d] \\
I \arrow[r, equal] & I \arrow[r, "u"] & J
\end{tikzcd}
$$

Similarly, there exists a unique morphism $\psi : X' \to X$ in $\mathbb{E}$ over I such that $f' = f \circ \psi$. Moreover, both $\mathrm{id}_{X'}$ and $\varphi \circ \psi$ are morphisms in $\mathbb{E}$ over I whose composition with f' is f' itself, as suggested in the following diagram.

$$\begin{tikzcd}
X' \arrow[dr, "\psi"] \arrow[drr, bend left, "f'"] \arrow[ddr, bend right, "\mathrm{id}_{X'}"'] \arrow[ddd] & & & \\
& X \arrow[dr, "\varphi"] \arrow[drr, bend left, "f"] \arrow[dd] & & \\
I \arrow[dr, equal] & & X' \arrow[r, "f'"] \arrow[dd] & Y \arrow[dd] \\
& I \arrow[dr, equal] & & \\
& & I \arrow[r, "u"] & J
\end{tikzcd}$$

Thus, since f' is Cartesian, $\varphi \circ \psi = \mathrm{id}_{X'}$. By symmetry, $\psi \circ \varphi = \mathrm{id}_X$, and hence φ is an isomorphism. $\square$

Solution, (ii). Suppose $\mathbb{E}$ is a displayed category over $\mathbb{B}$, $u : I \to J$ and $v : K \to I$ are morphisms in $\mathbb{B}$, $f : X \to Y$ is a Cartesian morphism in $\mathbb{E}$ over u, and $g, h : Z \to X$ are morphisms in $\mathbb{E}$ over v such that $f \circ g = f \circ h$. These suppositions are summarized in the following commutative diagram.

$$\begin{tikzcd}
Z \arrow[dr, bend left, "g"] \arrow[drr, bend left, "f\circ g = f\circ h"] \arrow[dr, bend right, "h"'] \arrow[dd] & & \\
& X \arrow[r, "f"] \arrow[dd] & Y \arrow[dd] \\
K \arrow[dr, "v"'] \arrow[drr, bend left, "u\circ v"] & & \\
& I \arrow[r, "u"] & J
\end{tikzcd}$$

Since f is Cartesian, this situation immediately implies that $g = h$. $\square$

1.1.2. Let $p : \mathbb{E} \to \mathbb{B}$ be a functor; assume $f : X \to Y$ is in $\mathbb{E}$ and put $u = pf$. Show that f is Cartesian if and only if for each $Z \in \mathbb{E}$ and $v : pZ \to pX$ in $\mathbb{B}$, the function

$$\mathbb{E}_v(Z, X) \xrightarrow{f\circ(-)} \mathbb{E}_{u\circ v}(Z, Y)$$

is an isomorphism.

Remark 1.1.2.1. We will prove this in the more general context of displayed categories (cf. [1]). However, this result is essentially an unpacking of the definition of Cartesianness and, in particular, of [1, Definition 5.1].

Solution. Suppose $\mathbb{E}$ is a displayed category over $\mathbb{B}$, $u : I \to J$ is a morphism in $\mathbb{B}$, and $f : X \to Y$ is a morphism in $\mathbb{E}$ over u.

($\Longrightarrow$) Suppose f is Cartesian, and take a morphism $v : K \to I$ in $\mathbb{B}$ and an object Z in $\mathbb{E}$ over K. Since f is Cartesian, this situation gives rise to a one-to-one correspondence between the dashed arrows in the diagram below.

$$\begin{tikzcd}
Z \arrow[d] \arrow[dr, dashed] \arrow[drr, dashed, bend left] & & \\
& X \arrow[r, "f"] \arrow[d] \arrow[dr, bend left, "u\circ v"] & Y \arrow[d] \\
K \arrow[r, "v"'] \arrow[urr, bend left] & I \arrow[r, "u"] & J
\end{tikzcd}$$

Indeed, given a morphism $g : Z \to X$ in $\mathbb{E}$ over v, composition with f yields a morphism $f \circ g : Z \to Y$ over $u \circ v$. Conversely, given a morphism $h : Z \to Y$ in $\mathbb{E}$ over $u \circ v$, there is a unique morphism $g : Z \to X$ in $\mathbb{E}$ over v such that $f \circ g = h$ since f is Cartesian. This is precisely the assertion that $\mathbb{E}_v(Z, X) \xrightarrow{f\circ(-)} \mathbb{E}_{u\circ v}(Z, Y)$ is an isomorphism (of sets).

($\Longleftarrow$) Conversely, suppose that $\mathbb{E}_v(Z, X) \xrightarrow{f\circ(-)} \mathbb{E}_{u\circ v}(Z, Y)$ is an isomorphism for every morphism $v : K \to I$ in $\mathbb{B}$ and an object Z in $\mathbb{E}$ over K. Consider a particular morphism $v : K \to I$ in $\mathbb{B}$ and an object Z in $\mathbb{E}$ over K, and suppose there is a morphism $h : Z \to Y$ in $\mathbb{E}$ over $u \circ v$. Then the assumed isomorphism implies that there is a unique morphism $g : Z \to X$ in $\mathbb{E}$ over v such that $h = f \circ g$, which shows that f is Cartesian. $\square$

1.1.3. Consider the total category of a fibration. Show that

(i) every morphism factors as a vertical map followed (diagrammatically) by a Cartesian one;

(ii) a Cartesian map above an isomorphism is an isomorphism. Espe-

cially a vertical Cartesian map is an isomorphism.

Solution, (i). Let $\mathbb{E}$ be a fibration over $\mathbb{B}$, and let $f : X \to Y$ be a morphism in $\mathbb{E}$ over a morphism $u : I \to J$ in $\mathbb{B}$. Then there exists a Cartesian morphism $f' : X' \to Y$ in $\mathbb{E}$ above u. Then we are in the situation of the diagram on the left below, so there exists a unique (vertical) morphism $g : X \to X'$ over I such that the diagram below on the right commutes.

$$\begin{tikzcd}
X \arrow[dd] \arrow[drr, bend left, "f"] & & \\
& X' \arrow[r, "f'"] \arrow[dd] \arrow[ddr, bend left, "u"] & Y \arrow[dd] \\
I \arrow[dr, equal] & & \\
& I \arrow[r, "u"] & J
\end{tikzcd}
\Longrightarrow
\begin{tikzcd}
X \arrow[dd] \arrow[dr, dashed, "g"] \arrow[drr, bend left, "f"] & & \\
& X' \arrow[r, "f'"] \arrow[dd] \arrow[ddr, bend left, "u"] & Y \arrow[dd] \\
I \arrow[dr, equal] & & \\
& I \arrow[r, "u"] & J
\end{tikzcd}$$

In particular, $f = f' \circ g$; in other words, f factors as a vertical morphism followed by a Cartesian morphism. $\square$

Solution, (ii). Let $\mathbb{E}$ be a fibration over $\mathbb{B}$, and let $f : X \to Y$ be a Cartesian morphism in $\mathbb{E}$ over an isomorphism $u : I \to J$ in $\mathbb{B}$. Let $v = u^{-1} : J \to I$ in $\mathbb{B}$, and consider the situation in the following commutative diagram.

$$\begin{tikzcd}
Y \arrow[dd] \arrow[dr, dashed, "\exists! g"] \arrow[drr, bend left, "\mathrm{id}_Y"] & & \\
& X \arrow[r, "f"] \arrow[dd] \arrow[ddr, bend left, "\mathrm{id}_J"] & Y \arrow[dd] \\
J \arrow[dr, "\sim", "v"'] & & \\
& I \arrow[r, "\sim", "u"'] & J
\end{tikzcd}$$

Here the dashed morphism $g : Y \to X$ exists uniquely by virtue of f being Cartesian. Since $f \circ g = \mathrm{id}_Y$ is Cartesian (by Exercise 1.1.4(i)) and f is Cartesian, Exercise 1.1.4(iii) implies that g is Cartesian over the isomorphism v, and $f \circ g = \mathrm{id}_Y$. The same argument then implies that there exists a morphism $h : X \to Y$ over $u = v^{-1}$ such that $g \circ h = \mathrm{id}_X$. But then

$$f = f \circ \mathrm{id}_X = f \circ g \circ h = \mathrm{id}_Y \circ h = h,$$

so $g \circ f = \mathrm{id}_X$, and hence $g = f^{-1}$, so f is an isomorphism. $\square$

1.1.4. Let $\begin{array}{c}\mathbb{E}\\ \downarrow p\\ \mathbb{B}\end{array}$ be a fibration. Prove that

(i) all isomorphisms in $\mathbb{E}$ are Cartesian;

(ii) if $\xrightarrow{f}\xrightarrow{g}$ is a composable pair of Cartesian morphisms in $\mathbb{E}$, then also their composite $\xrightarrow{g\circ f}$ is Cartesian.

Hence it makes sense to talk about the subcategory $\mathrm{Cart}(\mathbb{E}) \hookrightarrow \mathbb{E}$ having all objects from $\mathbb{E}$ but only the Cartesian arrows. We write $|p|$ for the composite $\mathrm{Cart}(\mathbb{E}) \hookrightarrow \mathbb{E} \to \mathbb{B}$.

(iii) Let $\xrightarrow{f}\xrightarrow{g}$ be a composable pair in $\mathbb{E}$ again. Show now that if g and $g \circ f$ are Cartesian, then f is Cartesian as well.

(iv) Verify that a consequence of (iii) is that the functor $|p| : \mathrm{Cart}(\mathbb{E}) \to \mathbb{B}$ is a fibration. From (ii) in the previous exercise it follows that all fibres of $|p|$ are groupoids (*i.e.* that all maps in the fibres are isomorphisms).

[This $|p|$ will be called the **fibration of objects** of p.]

Remark 1.1.4.1. We will prove this in the more general context of displayed categories (cf. [1]).

Solution, (i). Let $\mathbb{E}$ be a displayed category over $\mathbb{B}$ (not necessarily a fibration), and let $f : X \to Y$ be a displayed isomorphism in $\mathbb{E}$ above an isomorphism $u : I \to J$ in $\mathbb{B}$. Suppose $v : K \to I$ is a morphism in $\mathbb{B}$ and $g : Z \to Y$ is a morphism in $\mathbb{E}$ above $u \circ v : K \to J$. Then $f^{-1} \circ g : Z \to X$ is the unique morphism in $\mathbb{E}$ above v such that the following diagram commutes.

$$\begin{tikzcd}
Z \arrow[dr, dashed, "f^{-1}\circ g"] \arrow[drr, bend left, "g"] \arrow[dd] & & \\
& X \arrow[r, "f"] \arrow[dd] & Y \arrow[dd] \\
K \arrow[dr, "v"'] \arrow[drr, bend left, "u\circ v"] & & \\
& I \arrow[r, "u"] & J
\end{tikzcd}$$

Indeed, to verify uniqueness, suppose $h : Z \to X$ is another morphism in $\mathbb{E}$ above v such that $f \circ h = g$, and observe that

$$h = f^{-1} \circ f \circ h = f^{-1} \circ g.$$

Therefore f is Cartesian. $\square$

Remark 1.1.4.2. The conclusion of Exercise 1.1.4(ii) appears in the literature in, e.g., [3, Lemma 8.1.4(i)]

Solution, (ii). Suppose $\mathbb{E}$ is a displayed category over $\mathbb{B}$ (not necessarily a fibration), $u : I \to J$ and $v : J \to K$ are morphisms in $\mathbb{B}$, and $f : X \to Y$ and $g : Y \to Z$ are Cartesian morphisms in $\mathbb{E}$ over u and v, respectively. To show that $g \circ f : X \to Z$ is Cartesian over $v \circ u : I \to K$, consider a morphism $w : L \to I$ in $\mathbb{B}$ and a morphism $h : W \to Z$ in $\mathbb{E}$ above $v \circ u \circ w$. These data are summarized in the following commutative diagram.

$$\begin{tikzcd}
W \arrow[rrr, bend left, "h"] \arrow[d] & & & \\
 & X \arrow[r, "f"] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & Y \arrow[r, "g"] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & Z \arrow[d] \\
L \arrow[r, "w"] & I \arrow[r, "u"] & J \arrow[r, "v"] & K
\end{tikzcd}$$

Since g is Cartesian, there exists a unique morphism $h' : W \to Y$ in $\mathbb{E}$ above $u \circ w$ such that $h = g \circ h'$, as in the following commutative diagram.

$$\begin{tikzcd}
W \arrow[rrr, bend left, "h"] \arrow[rr, bend left, dashed, "h'"] \arrow[d] & & & \\
 & X \arrow[r, "f"] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & Y \arrow[r, "g"] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & Z \arrow[d] \\
L \arrow[r, "w"] & I \arrow[r, "u"] & J \arrow[r, "v"] & K
\end{tikzcd}$$

Moreover, since f is Cartesian, there exists a unique morphism $h'' : W \to X$

in $\mathbb{E}$ above w such that $h' = f \circ h''$, as in the following commutative diagram.

$$\begin{tikzcd}
W \arrow[dr, dashed, "h''"] \arrow[drr, bend left, "h'"] \arrow[drrr, bend left, "h"] \arrow[dd] & & & \\
& X \arrow[r, "f"] \arrow[dd] \arrow[dr, phantom, "\lrcorner"] & Y \arrow[r, "g"] \arrow[dd] \arrow[dr, phantom, "\lrcorner"] & Z \arrow[dd] \\
L \arrow[dr, "w"] & & & \\
& I \arrow[r, "u"] & J \arrow[r, "v"] & K
\end{tikzcd}$$

In particular, we have

$$h = g \circ h' = g \circ f \circ h''.$$

It remains to show that if $i : W \to X$ is another morphism in $\mathbb{E}$ above w such that $h = g \circ f \circ i$, then $i = h''$. Indeed, in this case the uniqueness of h' implies that $f \circ i = h'$, and so the uniqueness of h'' implies that $i = h''$. Thus, $g \circ f$ is Cartesian. $\square$

Solution, (iii). Suppose $\mathbb{E}$ is a displayed category over $\mathbb{B}$ (not necessarily a fibration), $u : I \to J$ and $v : J \to K$ are morphisms in $\mathbb{B}$, and $f : X \to Y$ and $g : Y \to Z$ are morphisms in $\mathbb{E}$ over u and v, respectively, such that g and $g \circ f$ are Cartesian. To show that $f : X \to Y$ is Cartesian over u, consider a morphism $w : L \to I$ in $\mathbb{B}$ and a morphism $h : W \to Y$ in $\mathbb{E}$ above $u \circ w$. This situation is summarized in the following commutative diagram.

$$\begin{tikzcd}
W \arrow[drrr, bend left, "h"] \arrow[dd] & & & \\
& X \arrow[r, "f"] \arrow[dd] & Y \arrow[r, "g"] \arrow[dd] & Z \arrow[dd] \\
L \arrow[dr, "w"] & & & \\
& I \arrow[r, "u"] & J \arrow[r, "v"] & K
\end{tikzcd}$$

But then $g \circ h : W \to Z$ is a morphism in $\mathbb{E}$ above $v \circ u \circ w$, so since $g \circ f$ is Cartesian, there exists a unique morphism $h' : W \to X$ in $\mathbb{E}$ over w such that $g \circ h = g \circ f \circ h'$. Since g is Cartesian, Exercise 1.1.1(i) implies that $h = f \circ h'$. Moreover, suppose $i : W \to X$ is another morphism above w such that $h = f \circ i$. Then $g \circ h = g \circ f \circ i$, so the uniqueness of h' arising from $g \circ f$ being Cartesian implies that $i = h'$. Thus, f is Cartesian. $\square$

Solution, (iv). Let $p : \mathbb{E} \to \mathbb{B}$ be a fibration; we will show that $|p| : \mathrm{Cart}(\mathbb{E}) \to \mathbb{B}$ is also a fibration. Consider a morphism $u : I \to J$ in $\mathbb{B}$ and an object Y in $\mathbb{E}$ over J. Since p is a fibration, there exists a p-Cartesian morphism $f : X \to Y$ in $\mathbb{E}$ over u. We claim that f is $|p|$-Cartesian as well. Indeed, suppose there is a morphism $v : K \to I$ in $\mathbb{B}$ and a p-Cartesian morphism $h : Z \to Y$ in $\mathbb{E}$ over $u \circ v$. Since f is p-Cartesian, there exists a unique morphism $h : Z \to X$ in $\mathbb{E}$ over v such that $g = f \circ h$. But now (iii) implies that h is p-Cartesian as well, so the entire diagram below is actually a diagram in the displayed category induced by $|p| : \mathrm{Cart}(\mathbb{E}) \to \mathbb{B}$.

$$\begin{tikzcd}
Z \arrow[dr, dashed, "\exists! h"] \arrow[drr, bend left, "g"] \arrow[dd] & & \\
& X \arrow[r, "f"] \arrow[dd] & Y \arrow[dd] \\
K \arrow[dr, "v"'] \arrow[drr, bend left, "u \circ v"] & & \\
& I \arrow[r, "u"] & J
\end{tikzcd}$$

Therefore, f is $|p|$-Cartesian, so $|p|$ is a fibration. $\square$

1.1.5. Verify that the following two results—known as the Pullback Lemmas—are a consequence of (ii) and (iii) in the previous exercise. Consider

$$\begin{tikzcd}
\bullet \arrow[r] \arrow[d] \arrow[dr, phantom, "(B)"] & \bullet \arrow[r] \arrow[d] \arrow[dr, phantom, "(A)"] & \bullet \arrow[d] \\
\bullet \arrow[r] & \bullet \arrow[r] & \bullet
\end{tikzcd}$$

(i) If (A) and (B) are pullback squares, then the outer rectangle is also a pullback square.

(ii) If the outer rectangle and (A) are pullback squares, then (B) is a pullback square as well.

Solution. Recall (e.g., from [4, Example 3.15] or [6, Proposition 1.1.6(ii)])

that in a category $\mathbb{B}$, a commutative square

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[d, "p"] & Y \arrow[d, "q"] \\ I \arrow[r, "u"] & J \end{tikzcd}$$

is a pullback square if and only if (u, f) is a Cartesian lift of u in the context of the codomain functor $\mathrm{cod} : \mathbb{B}^{\to} \to \mathbb{B}$, as in the following diagram.

$$\begin{tikzcd} \left(\begin{array}{c} X \\ \downarrow p \\ I \end{array}\right) \arrow[r, "(u,f)"] \arrow[d, "\mathrm{cod}"] & \left(\begin{array}{c} Y \\ \downarrow q \\ J \end{array}\right) \arrow[d, "\mathrm{cod}"] \\ I \arrow[r, "u"] & J \end{tikzcd}$$

Therefore, (i) and (ii) follow immediately from Exercise 1.1.4(ii) and Exercise 1.1.4(iii), respectively. □

1.1.6. Consider a functor $p : \mathbb{E} \to \mathbb{B}$. We describe a slightly weaker notion of Cartesianness, than the one above. Call a morphism $f : X \to Y$ in $\mathbb{E}$ **weak Cartesian** if for each $g : Z \to Y$ with $pf = pg$ there is a unique vertical $h : Z \to X$ with $f \circ h = g$. Show that the functor p is a fibration if and only if both

(a) every morphism $u : I \to pY$ in $\mathbb{B}$ has a weak Cartesian lifting $f : X \to Y$;

(b) the composition of two weak Cartesian morphisms is again weak Cartesian.

Remark 1.1.6.1. The conclusion of Exercise 1.1.6 appears in the literature in, e.g., [3, Proposition 8.1.7] and [5, Proposition 2.5].

Solution. ($\Longrightarrow$) Since every Cartesian morphism is weak Cartesian and compositions of Cartesian morphisms are Cartesian by Exercise 1.1.4(ii), every fibration satisfies (a) and (b).

($\Longrightarrow$) Conversely, suppose (a) and (b) hold. Let $u : I \to J$ be a morphism in $\mathbb{B}$, and let Y be an object in $\mathbb{E}$ above J. By (a), let $f : X \to Y$ be a weak Cartesian lift of u in $\mathbb{E}$. Consider another morphism $v : K \to I$ in $\mathbb{B}$ and a morphism $g : Z \to Y$ in $\mathbb{E}$ over $u \circ v$. By (a), choose a weak Cartesian lift $p : W \to X$ in $\mathbb{E}$ of v. Now $f \circ p$ is weak Cartesian over $u \circ v$ by (b), so there exists a unique vertical morphism $q : Z \to W$ over K such that the following diagram commutes.

$$\begin{tikzcd}
Z \arrow[dr, dashed, "q"] \arrow[drrr, bend left, "g"] \arrow[dd] & & & \\
& W \arrow[r, "p"] \arrow[dd] & X \arrow[r, "f"] \arrow[dd] & Y \arrow[dd] \\
K \arrow[dr, equal] & & & \\
& K \arrow[r, "v"] & I \arrow[r, "u"] & J
\end{tikzcd}$$

In particular, the following diagram commutes.

$$\begin{tikzcd}
Z \arrow[dr, "p\circ q"] \arrow[drrr, bend left, "g"] \arrow[dd] & & & \\
& X \arrow[r, "f"] \arrow[dd] & Y \arrow[dd] \\
K \arrow[dr, "v"] & & & \\
& I \arrow[r, "u"] & J
\end{tikzcd}$$

It remains to show that if $h : Z \to X$ is another morphism in $\mathbb{E}$ above v such that $g = f \circ h$, then $h = p \circ q$. In this case, since p is weak Cartesian, we can find a unique vertical morphism $q' : Z \to W$ over K such that $h = p \circ q'$.

$$\begin{tikzcd}
Z \arrow[dr, dashed, "q'"] \arrow[drr, bend left, "h"] \arrow[drrr, bend left, "g"] \arrow[dd] & & & \\
& W \arrow[r, "p"] \arrow[dd] & X \arrow[r, "f"] \arrow[dd] & Y \arrow[dd] \\
K \arrow[dr, equal] & & & \\
& K \arrow[r, "v"] & I \arrow[r, "u"] & J
\end{tikzcd}$$

In particular, $g = f \circ p \circ q'$, so the uniqueness of q implies that $q' = q$, and hence $h = p \circ q$. Thus, f is a Cartesian lift of u, so p is a fibration. $\square$

1.1.7. Check that the following are (trivial) examples of fibrations

$$\begin{tikzcd} \mathbb{B} \times \mathbb{C} \arrow[d, "\mathrm{fst}"] \\ \mathbb{B} \end{tikzcd} \qquad \begin{tikzcd} \mathbb{B} \arrow[d, "\mathrm{id}"] \\ \mathbb{B} \end{tikzcd} \qquad \begin{tikzcd} \mathbb{B} \arrow[d] \\ 1 = \{*\} \end{tikzcd} \qquad \begin{tikzcd} X \arrow[d] \\ I \end{tikzcd}$$

where X, I are sets (*i.e.* discrete categories).

Solution. Consider the functor $\mathbb{B} \times \mathbb{C} \xrightarrow{\mathrm{fst}} \mathbb{B}$, which projects onto the first component. Consider a morphism $u : I \to J$ in $\mathbb{B}$ and an object (J, Y) in $\mathbb{B} \times \mathbb{C}$ above J. We claim that $(v, \mathrm{id}_Y) : (I, Y) \to (J, Y)$ is a Cartesian lift of u. Suppose there are morphisms $v : K \to J$ in $\mathbb{B}$ and $(w, g) : (K, Z) \to (J, Y)$ in $\mathbb{B} \times \mathbb{C}$ above $u \circ v$ (i.e., $w = u \circ v$), as in the following diagram.

$$\begin{tikzcd} (K,Z) \arrow[dr, dashed, "?"] \arrow[drr, bend left, "(w,g)"] \arrow[dd, "\mathrm{fst}"] & & \\ & (I,Y) \arrow[r, "(u,\mathrm{id}_Y)"] \arrow[d, "\mathrm{fst}"] & (J,Y) \arrow[d, "\mathrm{fst}"] \\ K \arrow[r, "v"] & I \arrow[r, "u"] & J \end{tikzcd}$$

Observe that $(v, g) : (K, Z) \to (I, Y)$ is the unique morphism in $\mathbb{B} \times \mathbb{C}$ that fills in the dashed "?" morphism making the diagram above commute, so (u, id_Y) is Cartesian and hence $\mathbb{B} \times \mathbb{C} \xrightarrow{\mathrm{fst}} \mathbb{B}$ is a fibration.

Next, consider the identity functor $\mathbb{B} \xrightarrow{\mathrm{id}} \mathbb{B}$, and observe that this situation is equivalent to the functor $\mathbb{B} \times 1 \xrightarrow{\mathrm{fst}} \mathbb{B}$, so it is a fibration by the first part of this exercise. More explicitly, consider a morphism $u : I \to J$ in $\mathbb{B}$. The only object above J in this scenario is J itself, and we claim that u itself is the unique Carteisan lift of u. Indeed, given morphisms $v : K \to I$ in $\mathbb{B}$ ("below") and $w : K \to J$ in $\mathbb{B}$ ("above") over $u \circ v$ (i.e., $w = u \circ v$), the

unique morphism $K \to I$ which fills in the dashed "?" arrow in the diagram below is v itself.

$$\begin{tikzcd} K \arrow[rr, bend left, "w"] \arrow[dr, dashed, bend left, "?"] \arrow[dr, bend right, "v"'] & & J \\ & I \arrow[ur, "u"] & \end{tikzcd}$$

Thus, u is a Cartesian lift of u, and $\mathbb{B} \xrightarrow{\mathrm{id}} \mathbb{B}$ is a fibration.

Consider the functor $\mathbb{B} \to 1$, and observe that this situation is equivalent to the functor $1 \times \mathbb{B} \xrightarrow{\mathrm{fst}} 1$, so it is a fibration by the first part of this exercise. More explicitly, consider *the* morphism $* \to *$ in 1, and let Y be an object in $\mathbb{B}$ (vacuously above $*$). Then $\mathrm{id}_Y : Y \to Y$ is a Cartesian lift of $* \to *$, whence $\mathbb{B} \to 1$ is a fibration.

Finally, let X and Y be sets and $p : X \to I$ a function, considered as a functor between discrete categories. All morphisms in I and X are identities, so if $i \in I$ and $x \in X$ is a "lift" of i (i.e., $p(x) = i$ or equivalently $x \in p^{-1}(i)$), then id_x is a Cartesian lift of id_i. Therefore, $p : X \to I$ is a fibration. $\square$

1.1.8. For an arbitrary category $\mathbb{B}$, consider the domain functor $\mathrm{dom} : \mathbb{B}^{\to} \to \mathbb{B}$.

(i) Describe the fibre category above $I \in \mathbb{B}$. It is usually called the **opslice category** or simply **opslice** and written as $I \setminus \mathbb{B}$.

(ii) Show that dom is a fibration (without any assumptions about $\mathbb{B}$).

(iii) Show also that for each $I \in \mathbb{B}$ the domain functor $\mathrm{dom}_I : \mathbb{B}/I \to \mathbb{B}$ is a fibration.

Solution, (i). For a category $\mathbb{B}$ and an object $I \in \mathbb{B}$, the category $I \setminus \mathbb{B}$ has

objects Morphisms $\begin{pmatrix} I \\ \downarrow\varphi \\ X \end{pmatrix}$ in $\mathbb{B}$ with domain I.

morphisms Given objects $\begin{pmatrix} I \\ \downarrow\varphi \\ X \end{pmatrix}$ and $\begin{pmatrix} I \\ \downarrow\psi \\ Y \end{pmatrix}$ in $I \setminus \mathbb{B}$, a morphism in $I \setminus \mathbb{B}$ from the first to the second consists of a morphism $f : X \to Y$ in $\mathbb{B}$

such that the following diagram commutes.

$$\begin{tikzcd} & I \arrow[dl, "\varphi"'] \arrow[dr, "\psi"] & \\ X \arrow[rr, "f"] & & Y \end{tikzcd}$$

□

Solution, (ii). Consider a morphism $u : I \to J$ in $\mathbb{B}$, and another morphism $\psi : J \to Y$ in $\mathbb{B}$ (i.e., an object $\begin{pmatrix} J \\ \downarrow\psi \\ Y \end{pmatrix}$ in $\mathbb{B}^{\to}$ over J with respect to $\mathrm{dom} : \mathbb{B}^{\to} \to \mathbb{B}$). We claim that

$$\begin{pmatrix} I \\ \downarrow \psi \circ u \\ Y \end{pmatrix} \xrightarrow{(\mathrm{id}_Y, u)} \begin{pmatrix} J \\ \downarrow \psi \\ Y \end{pmatrix} \qquad \text{i.e.,} \qquad \begin{tikzcd} I \arrow[r, "u"] \arrow[d, "\psi\circ u"'] & J \arrow[d, "\psi"] \\ Y \arrow[r, equal] & Y \end{tikzcd} \tag{1.1.8.1}$$

is a Cartesian lift of u. Indeed, consider a situation as in the following commutative diagram of solid arrows.

$$\begin{tikzcd} \begin{pmatrix} K \\ \downarrow \varphi \\ Z \end{pmatrix} \arrow[rr, bend left, "(g, u\circ v)"] \arrow[dr, dashed, "?"] \arrow[dd, "\mathrm{dom}"'] & & \\ & \begin{pmatrix} I \\ \downarrow \psi \circ u \\ Y \end{pmatrix} \arrow[r, "(\mathrm{id}_Y, u)"] \arrow[dd, "\mathrm{dom}"] & \begin{pmatrix} J \\ \downarrow \psi \\ Y \end{pmatrix} \arrow[dd, "\mathrm{dom}"] \\ K \arrow[dr, "v"'] & & \\ & I \arrow[r, "u"] & J \end{tikzcd} \tag{1.1.8.2}$$

We must find a unique morphism in $\mathbb{B}^{\to}$ to fill in the dashed "?" arrow in (1.1.8.2) that makes the diagram commute. But observe that (1.1.8.2) is

equivalent to the following diagram.

$$\begin{tikzcd}
K \arrow[r, "v"] \arrow[d, "\varphi"] & I \arrow[r, "u"] \arrow[d, "\psi\circ u"] & J \arrow[d, "\psi"] \\
Z \arrow[r, dashed, "?"] \arrow[rr, bend right, "g"'] & Y \arrow[r, equal] & Y
\end{tikzcd}$$

It becomes evident that the unique morphism that fills in the dashed "?" arrow in (1.1.8.2) to makes the diagram commute is $? = (g, v)$. Thus, (1.1.8.1) is a Cartesian lift of u, and hence $\mathrm{dom} : \mathbb{B}^{\to} \to \mathbb{B}$ is a fibration. $\square$

Solution, (iii). The same proof as in (ii) applies here as well. Explicitly, fix an object I in $\mathbb{B}$, let $u : X \to Y$ be a morphism in $\mathbb{B}$, and let $\begin{pmatrix} Y \\ \downarrow\psi \\ I \end{pmatrix}$ in $\mathbb{B}/I$ over Y relative to the functor $\mathrm{dom}_I : \mathbb{B}/I \to \mathbb{B}$. Then a straightforward modification of the proof of (ii) shows that

$$\begin{pmatrix} X \\ \downarrow \psi\circ u \\ I \end{pmatrix} \xrightarrow{u} \begin{pmatrix} Y \\ \downarrow \psi \\ I \end{pmatrix} \qquad \text{i.e.,} \qquad \begin{tikzcd}
X \arrow[rr, "u"] \arrow[dr, "\psi\circ u"'] & & J \arrow[dl, "\psi"] \\
& I &
\end{tikzcd}$$

is a Cartesian lift of u, and hence $\mathrm{dom}_I : \mathbb{B}/I \to \mathbb{B}$ is a fibration. $\square$

1.1.9. Assume $\mathbb{B}$ is a category with pullbacks. Show that the functor $\mathbb{B}^{\to\to} \to \mathbb{B}^{\to}$ sending $\xrightarrow{f}\xrightarrow{g}$ to $\xrightarrow{g}$ is a fibration. Is the composite $B^{\to\to} \to \mathbb{B}^{\to} \to \mathbb{B}$ also a fibration?

Solution. Let

$$\begin{pmatrix} X \\ \downarrow \varphi \\ I \end{pmatrix} \xrightarrow{(u,f)} \begin{pmatrix} Y \\ \downarrow \psi \\ J \end{pmatrix} \qquad \text{i.e.,} \qquad \begin{tikzcd}
X \arrow[r, "f"] \arrow[d, "\varphi"'] & Y \arrow[d, "\psi"] \\
I \arrow[r, "u"] & J
\end{tikzcd}$$

be a morphism in $\mathbb{B}^{\to}$, and let $\begin{pmatrix} B \\ \downarrow\beta \\ Y \\ \downarrow\psi \\ I \end{pmatrix}$ be an object in $\mathbb{B}^{\to\to}$ "above" $\begin{pmatrix} Y \\ \downarrow\psi \\ I \end{pmatrix}$.

Since $\mathbb{B}$ has pullbacks, let $X \xleftarrow{\alpha} A \xrightarrow{q} B$ be a pullback of $X \xrightarrow{f} Y \xleftarrow{\beta} B$. We claim that

$$\begin{pmatrix} A \\ \downarrow\alpha \\ X \\ \downarrow\varphi \\ I \end{pmatrix} \xrightarrow{(u,f,q)} \begin{pmatrix} B \\ \downarrow\beta \\ Y \\ \downarrow\psi \\ J \end{pmatrix} \qquad \text{i.e.,} \qquad \begin{tikzcd} A \arrow[r, "q"] \arrow[d, "\alpha"'] \arrow[dr, phantom, "\lrcorner"] & B \arrow[d, "\beta"] \\ X \arrow[r, "f"] \arrow[d, "\varphi"'] & Y \arrow[d, "\psi"] \\ I \arrow[r, "u"] & J \end{tikzcd}$$

is a Cartesian lift of (u, f). Indeed, consider a situation as in the following diagram.

$$\begin{tikzcd} \begin{pmatrix} C \\ \downarrow\gamma \\ Z \\ \downarrow\xi \\ K \end{pmatrix} \arrow[dr, dashed, "?"] \arrow[drr, bend left, "(w,h,r)"] \arrow[dd] & & \\ & \begin{pmatrix} A \\ \downarrow\alpha \\ X \\ \downarrow\varphi \\ I \end{pmatrix} \arrow[r, "(u,f,q)"] \arrow[dd] & \begin{pmatrix} B \\ \downarrow\beta \\ Y \\ \downarrow\psi \\ J \end{pmatrix} \arrow[dd] \\ \begin{pmatrix} Z \\ \downarrow\xi \\ K \end{pmatrix} \arrow[dr, "(v,g)"'] & & \\ & \begin{pmatrix} X \\ \downarrow\varphi \\ I \end{pmatrix} \arrow[r, "(u,f)"] & \begin{pmatrix} Y \\ \downarrow\psi \\ J \end{pmatrix} \end{tikzcd} \tag{1.1.9.1}$$

(where necessarily $w = u \circ v$ and $h = f \circ g$). We must find a unique filler for the dashed morphism labeled "?"; note that its first and second components

are necessarily v and g, respectively. Also note that the diagram above is equivalent to the following one.

$$\begin{tikzcd}
C \arrow[r, dashed, "p"] \arrow[rr, bend left, "r"] \arrow[d, "\gamma"] & A \arrow[r, "q"] \arrow[d, "\alpha"] \arrow[dr, phantom, "\lrcorner"] & B \arrow[d, "\beta"] \\
Z \arrow[r, "g"] \arrow[rr, bend right, "h"'] \arrow[d, "\xi"] & X \arrow[r, "f"] \arrow[d, "\varphi"] & Y \arrow[d, "\psi"] \\
K \arrow[r, "v"] \arrow[rr, bend right, "w"'] & I \arrow[r, "u"] & J
\end{tikzcd}$$

Here the dashed arrow $p : C \to A$ is the unique morphism in $\mathbb{B}$ such that $g \circ \gamma = \alpha \circ p$ and $r = q \circ p$, by virtue of $X \xleftarrow{\alpha} A \xrightarrow{q} B$ being a pullback of $X \xrightarrow{f} Y \xleftarrow{\beta} B$. It follows that the unique filler for "?" in (1.1.9.1) is $? = (v, g, p)$, so (u, f, q) is a Cartesian lift of (u, f), and hence the functor $\mathbb{B}^{\to\to} \to \mathbb{B}^{\to}$ sending $\xrightarrow{f}\xrightarrow{g}$ to $\xrightarrow{g}$ is a fibration.

Moreover, the composite $B^{\to\to} \to \mathbb{B}^{\to} \to \mathbb{B}$ is also a fibration since the composition of two fibrations is again a fibration (cf. [3, Proposition 8.1.12], [4, Proposition 3.7], [5, Proposition 3.1] [7, Proposition 9.1.10]) and since the codomain functor $\mathrm{cod} : \mathbb{B}^{\to} \to \mathbb{B}$ is a fibration when $\mathbb{B}$ has pullbacks (cf. [6, Proposition 1.1.6]). $\square$

1.1.10. Let **Fld** be the category of fields and field homomorphisms; **Vect** is the category of vector spaces: objects are triples $(K, V, \cdot)$ where K is a field of scalars, V is an Abelian group of vectors and $\cdot : K \times V \to V$ is an action of scalar multiplication (which distributes both over scalar and over vector addition). A morphism $(K, V, \cdot) \to (L, W, \cdot)$ in **Vect** is a pair (u, f) where $u : K \to L$ is a field homomorphism and $f : V \to W$ a group homomorphism such that $f(a \cdot x) = u(a) \cdot f(x)$ for all $a \in K$ and $x \in V$.

Check that the obvious forgetful functor **Vect** $\to$ **Fld** is a fibration. What are the fibres? Which maps are Cartesian?

Solution. First, the fibres of **Vect** $\to$ **Fld** are the categories of vector spaces over a fixed field K, with morphisms being ordinary K-linear maps.

Next, we show that **Vect** $\to$ **Fld** is a fibration. Let $u : K \to L$ be a field homomorphism, and let $(L, V, \cdot)$ be an object of **Vect** over L. Then we claim that (u, id_V) is a Cartesian lift of u. Indeed, suppose $v : F \to K$ is another field homomorphism and $(w, g) : (F, W, \cdot) \to (L, V, \cdot)$ be a morphism in **Vect** over $u \circ v$ such that the following diagram commutes.

$$\begin{tikzcd}
(F, W, \cdot) \arrow[dr, dashed, "?"] \arrow[drr, bend left, "(w,g)"] \arrow[dd] & & \\
& (K, V, \cdot) \arrow[r, "(u,\mathrm{id}_V)"] \arrow[d] & (L, V, \cdot) \arrow[d] \\
F \arrow[r, "v"] & K \arrow[r, "u"] & L
\end{tikzcd}$$

The unique filler of the dashed "?" morphism in this diagram is $? = (v, g)$, so (u, id_V) is Cartesian, and hence **Vect** $\to$ **Fld** is Cartesian.

Finally, we claim that a morphism $(u, f) : (K, V, \cdot) \to (L, W, \cdot)$ is Cartesian if and only if $f : V \to W$ is an abelian group isomorphism. To show that abelian group isomorphisms induce Cartesian morphisms, mimic the proof above that showed that (u, id_V) is a Cartesian lift of $u : K \to L$ whenever $(L, V, \cdot)$ is an object in **Vect** above L. Conversely, suppose $(u, f) : (K, V, \cdot) \to (L, W, \cdot)$ is a Cartesian morphism in **Vect** above a field homomorphism $u : K \to L$. We claim that $f : V \to W$ is an abelian group isomorphism. We already showed that in this case $(u, \mathrm{id}_W) : (K, W, \cdot) \to (L, W, \cdot)$ is a Cartesian lift of u, so by Exercise 1.1.1(i), there exists a unique isomorphism $(w, g) : (K, V, \cdot) \to (K, W, \cdot)$ in **Vect** such that $(u, f) = (u, \mathrm{id}_W) \circ (w, g)$. In particular, g is an abelian group isomorphism and $f = \mathrm{id}_W \circ g = g$, so f is an abelian group isomorphism. $\square$

1.2 Some concrete examples: sets, ω-sets and PERs

1.2.1. Prove that a morphism $\left(u, (f_i)_{i \in I}\right)$ in $\mathrm{Fam}(\mathbb{C})$ is Cartesian if and only if each f_i is an isomorphism in $\mathbb{C}$.

Solution. ($\Longrightarrow$) Suppose

$$\left(u, (f_i)_{i\in I}\right) : (X_i)_{i\in I} \longrightarrow (Y_j)_{j\in J}$$

is a Cartesian morphism in $\mathrm{Fam}(\mathbb{C})$ over a morphism $u : I \to J$ in $\mathbb{C}$. We also know that

$$\left(u, \left(\mathrm{id}_{Y_{u(i)}}\right)_{i\in I}\right) : \left(Y_{u(i)}\right)_{i\in I} \longrightarrow (Y_j)_{j\in J}$$

is another Cartesian lift of u (cf. [6, preceding Definition 1.2.1]). Thus, by Exercise 1.1.1(i), there is a unique vertical isomorphism

$$\left(\mathrm{id}_I, \left(X_i \xrightarrow{g_i} Y_{u(i)}\right)_{i\in I}\right) : (X_i)_{i\in I} \longrightarrow \left(Y_{u(i)}\right)_{i\in I}$$

such that

$$\left(u, (f_i)_{i\in I}\right) = \left(u, \left(\mathrm{id}_{Y_{u(i)}}\right)_{i\in I}\right) \circ \left(\mathrm{id}_I, (g_i)_{i\in I}\right)$$

in $\mathrm{Fam}(\mathbb{C})$. In particular, $f_i = \mathrm{id}_{Y_{u(i)}} \circ g_i = g_i$ for all $i \in I$, so each f_i is an isomorphism.

($\Longleftarrow$) Conversely, suppose

$$\left(u, (f_i)_{i\in I}\right) : (X_i)_{i\in I} \longrightarrow (Y_j)_{j\in J}$$

is a morphism in $\mathrm{Fam}(\mathbb{C})$ over a morphism $u : I \to J$ in $\mathbb{C}$ such that each $f_i : X_i \to Y_{u(i)}$ is an isomorphism. We claim that $\left(u, (f_i)_{i\in I}\right)$ is Cartesian. Suppose $v : K \to I$ is another morphism in $\mathbb{C}$ and

$$\left(u \circ v, (g_k)_{k\in K}\right) : (Z_k)_{k\in K} \longrightarrow (Y_j)_{j\in J}$$

is a morphism in $\mathrm{Fam}(\mathbb{C})$ over $u \circ v$, as in the following diagram.

$$\begin{tikzcd}
(Z_k)_{k\in K} \arrow[dr, dashed, "?"] \arrow[drr, bend left, "\left(u\circ v, (g_k)_{k\in K}\right)"] \arrow[dd] & & \\
& (X_i)_{i\in I} \arrow[r, "\left(u, (f_i)_{i\in I}\right)"] \arrow[d] & (Y_j)_{j\in J} \arrow[d] \\
K \arrow[dr, "v"'] & I \arrow[r, "u"] & J \\
& &
\end{tikzcd}$$

Then the unique filler of the dashed "?" morphism in this diagram is $? = \left(v, \left(f_{v(k)}^{-1} \circ g_k\right)_{k\in K}\right)$, so $\left(u, (f_i)_{i\in I}\right)$ is Cartesian. $\square$

1.2.2. Consider a map $f = (f_i : X_i \to Y_i)_{i \in I}$ in the fibre $\mathrm{Fam}(\mathbb{C})_I = \mathbb{C}^I$ over $I \in \mathbf{Sets}$. Prove that f is a mono in this fibre if and only if each f_i is a mono in $\mathbb{C}$.

Solution. ($\Longrightarrow$) Suppose $f = (f_i : X_i \to Y_i)_{i \in I}$ is monic in $\mathbb{C}^I$, and fix $i \in I$. Suppose $g, h : Z \to X_i$ are arrows in $\mathbb{C}$ such that $f_i \circ g = f_i \circ h$. We claim that $h = g$, which will show that f_i is monic. Define a family of object $(Z_j)_{j \in I}$ in $\mathbb{C}^I$, where

$$Z_j = \begin{cases} Z & \text{if } i = j \\ X_j & \text{otherwise,} \end{cases}$$

and define morphisms $g' = \left(g'_j : Z_j \to X_j\right)_{j \in I} : (Z_j)_{j \in I} \to (X_j)_{j \in I}$ and $h' = \left(h'_j : Z_j \to X_j\right)_{j \in I} : (Z_j)_{j \in I} \to (X_j)_{j \in I}$ in $\mathbb{C}^I$, where

$$g'_j = \begin{cases} g & \text{if } i = j \\ \mathrm{id}_{X_j} & \text{otherwise,} \end{cases} \qquad h'_j = \begin{cases} h & \text{if } i = j \\ \mathrm{id}_{X_j} & \text{otherwise,} \end{cases}$$

Then we have $f \circ g' = f \circ h'$, so $g' = h'$, and in particular $g = g'_i = h'_i = h$.

($\Longleftarrow$) Suppose $f = (f_i : X_i \to Y_i)_{i \in I}$ is a morphism in $\mathbb{C}^I$ and f_i is monic in $\mathbb{C}$ for each $i \in I$. Consider morphisms $g = (g_j : Z_j \to X_j)_{j \in I} : (Z_j)_{j \in I} \to (X_j)_{j \in I}$ and $h = (h_j : Z_j \to X_j)_{j \in I} : (Z_j)_{j \in I} \to (X_j)_{j \in I}$ in $\mathbb{C}^I$ such that $f \circ g = f \circ h$. In particular, $f_i \circ g_i = f_i \circ h_i$ for each $i \in I$, so $g_i = h_i$ (since f_i is monic) for each $i \in I$. Therefore, $g = h$, and hence f is monic. $\square$

1.2.3. For an arbitrary category $\mathbb{B}$, let $\mathbb{B}^{\to}_{\bullet}$ be the category with **pointed families** as objects; these are pairs $\left\langle \begin{pmatrix} X \\ \downarrow\varphi \\ I \end{pmatrix}, s \right\rangle$ where s is a section of φ (*i.e.* a map $s : I \to X$ with $\varphi \circ s = \mathrm{id}$). A morphism $\left\langle \begin{pmatrix} X \\ \downarrow\varphi \\ I \end{pmatrix}, s \right\rangle \to \left\langle \begin{pmatrix} Y \\ \downarrow\psi \\ J \end{pmatrix}, t \right\rangle$ in $\mathbb{B}^{\to}_{\bullet}$ consists of a pair of morphisms $u : I \to J$, $f : X \to Y$ in $\mathbb{B}$ with $\psi \circ f = u \circ \varphi$ and also $f \circ s = t \circ u$. Thus morphisms of pointed

families preserve the points (*i.e.* sections) of the families. Prove that

(i) if the category $\mathbb{B}$ has pullbacks then the functor $\mathbb{B}^{\to}_{\bullet}$ sending $\left\langle \begin{pmatrix} X \\ \downarrow\varphi \\ I \end{pmatrix}, s \right\rangle$ to the index set I is a fibration;

(ii) for $\mathbb{B} = \mathbf{Sets}$, there is an equivalence of categories $\mathrm{Fam}(\mathbf{Sets}_\bullet) \xrightarrow{\simeq} \mathbf{Sets}^{\to}_{\bullet}$ like in Proposition 1.2.2, where $\mathbf{Sets}_\bullet$ is the category of **pointed sets**: objects are sets containing a distinguished base point, morphisms are functions preserving such points.

Solution, (i). Let $\mathbb{B}$ be a category with pullbacks, and consider the functor

$$\begin{tikzcd}
\mathbb{B}^{\to}_{\bullet} \arrow[r] & \mathbb{B} \\
\left\langle \begin{pmatrix} X \\ \downarrow\varphi \\ I \end{pmatrix}, s \right\rangle \arrow[r, mapsto] \arrow[d, "{(u,f)}"', mapsto] & I \arrow[d, "u"] \\
\left\langle \begin{pmatrix} Y \\ \downarrow\psi \\ J \end{pmatrix}, t \right\rangle \arrow[r, mapsto] & J
\end{tikzcd}$$

where the morphism (u, f) represents the commutative diagram

$$\begin{tikzcd}
I \arrow[r, "u"] \arrow[d, "s"] \arrow[dd, bend right=60, "\mathrm{id}_I"'] & J \arrow[d, "t"'] \arrow[dd, bend left=60, "\mathrm{id}_J"] \\
X \arrow[r, "f"] \arrow[d, "\varphi"] & Y \arrow[d, "\psi"'] \\
I \arrow[r, "u"] & J
\end{tikzcd} \tag{1.2.3.1}$$

We claim that $\mathbb{B}^{\to}_{\bullet} \to \mathbb{B}$ is a fibration. To this end, let $u : I \to J$ be a

morphism in $\mathbb{B}$, and let $\left\langle \begin{pmatrix} Y \\ \downarrow\psi \\ J \end{pmatrix}, t \right\rangle$ be an object in $\mathbb{B}^{\to}_{\bullet}$ above J. Since $\mathbb{B}$ has pullbacks, let $I \xleftarrow{\varphi} X \xrightarrow{f} Y$ be a pullback of $I \xrightarrow{u} J \xleftarrow{\psi} Y$. Moreover, define $s : I \to X$ as the unique dashed arrow that makes the following diagram commute, by virtue of $I \xleftarrow{\varphi} X \xrightarrow{f} Y$ being a pullback of $I \xrightarrow{u} J \xleftarrow{\psi} Y$.

$$\begin{tikzcd}
I \arrow[r, "u"] \arrow[dr, dashed, "s"] \arrow[ddr, bend right, "\mathrm{id}_I"'] & J \arrow[dr, "t"] & \\
& X \arrow[r, "f"] \arrow[d, "\varphi"] \arrow[dr, phantom, "\lrcorner", very near start] & Y \arrow[d, "\psi"] \\
& I \arrow[r, "u"] & J
\end{tikzcd}$$

In particular, $\varphi \circ s = \mathrm{id}_I$, so $\left\langle \begin{pmatrix} X \\ \downarrow\varphi \\ I \end{pmatrix}, s \right\rangle$ is an object in $\mathbb{B}^{\to}_{\bullet}$. Also, we have a situation as in (1.2.3.1), namely that

$$\psi \circ f = u \circ \varphi, \qquad\qquad f \circ s = t \circ u,$$

so we have a morphism

$$\left\langle \begin{pmatrix} X \\ \downarrow\varphi \\ I \end{pmatrix}, s \right\rangle \xrightarrow{(u,f)} \left\langle \begin{pmatrix} Y \\ \downarrow\psi \\ J \end{pmatrix}, t \right\rangle \tag{1.2.3.2}$$

in $\mathbb{B}^{\to}_{\bullet}$. We claim that it is a Cartesian lift of u. Indeed, consider a morphism $v : K \to I$ in $\mathbb{B}$ and a morphism

$$\left\langle \begin{pmatrix} Z \\ \downarrow\xi \\ K \end{pmatrix}, r \right\rangle \xrightarrow{(u\circ v,g)} \left\langle \begin{pmatrix} Y \\ \downarrow\psi \\ J \end{pmatrix}, t \right\rangle$$

in $\mathbb{B}^{\to}_{\bullet}$ above $u \circ v$. These data are summarized by the solid arrows in the

following commutative diagram.

$$\begin{tikzcd}
\left\langle \begin{pmatrix} Z \\ \downarrow \xi \\ K \end{pmatrix}, r \right\rangle \arrow[rr, bend left, "(u\circ v, g)"] \arrow[r, dashed, "?"] \arrow[d] & \left\langle \begin{pmatrix} X \\ \downarrow \varphi \\ I \end{pmatrix}, s \right\rangle \arrow[r, "(u,f)"] \arrow[d] & \left\langle \begin{pmatrix} Y \\ \downarrow \psi \\ J \end{pmatrix}, t \right\rangle \arrow[d] \\
K \arrow[r, "v"] & I \arrow[r, "u"] & J
\end{tikzcd} \tag{1.2.3.3}$$

We must find a unique filler for the dashed arrow above v labeled "?" in (1.2.3.3) that makes the diagram commute. Moreover, the situation in (1.2.3.3) can also be summarized by the following commutative diagram in $\mathbb{B}$.

$$\begin{tikzcd}
K \arrow[r, "v"] \arrow[d, "r"] \arrow[dd, bend right=60, "\mathrm{id}_K"'] & I \arrow[r, "u"] \arrow[d, "s"] \arrow[dd, bend right=60, "\mathrm{id}_I"'] & J \arrow[d, "t"'] \arrow[dd, bend left=60, "\mathrm{id}_J"] \\
Z \arrow[d, "\xi"] \arrow[urr, bend left] & X \arrow[r, "f"] \arrow[d, "\varphi"] \arrow[urr, bend left, "g"] \arrow[dr, phantom, "\lrcorner"] & Y \arrow[d, "\psi"] \\
K \arrow[r, "v"] & I \arrow[r, "u"] & J
\end{tikzcd} \tag{1.2.3.4}$$

In particular, since $\psi \circ g = u \circ v \circ \xi$, the universal property of the pullback $I \xleftarrow{\varphi} X \xrightarrow{f} Y$ implies that there exists a unique morphism $h : Z \to X$ in $\mathbb{B}$ such that the following diagram in $\mathbb{B}$ commutes.

$$\begin{tikzcd}
Z \arrow[d, "\xi"'] \arrow[dr, dashed, "h"] \arrow[drr, bend left, "g"] & & \\
K \arrow[dr, "v"'] & X \arrow[r, "f"] \arrow[d, "\varphi"'] \arrow[dr, phantom, "\lrcorner"] & Y \arrow[d, "\psi"] \\
& I \arrow[r, "u"] & J
\end{tikzcd}$$

Also, since we have both

$$\begin{aligned}&\psi \circ f \circ s \circ v = u \circ \varphi \circ s \circ v,\\ &\psi \circ f \circ h \circ r = \psi \circ g \circ r = u \circ v \circ \xi \circ r = u \circ \varphi \circ h \circ r,\end{aligned}$$

the universal property of the pullback $I \xleftarrow{\varphi} X \xrightarrow{f} Y$ implies that $s \circ v = h \circ r$, as in the following commutative diagram in $\mathbb{B}$.

$$\begin{tikzcd}
K \arrow[dr, bend left, "s\circ v"] \arrow[dr, bend right, "h\circ r"'] \arrow[dr, phantom, "/\!/"] & & \\
& X \arrow[r, "f"] \arrow[d, "\varphi"'] \arrow[dr, phantom, "\lrcorner"] & Y \arrow[d, "\psi"] \\
& I \arrow[r, "u"] & J
\end{tikzcd}$$

Thus, (1.2.3.4) can be extended into the following commutative diagram.

$$\begin{tikzcd}
K \arrow[r, "v"] \arrow[d, "r"] \arrow[dd, bend right=60, "\mathrm{id}_K"'] & I \arrow[r, "u"] \arrow[d, "s"] \arrow[dd, bend right=60, "\mathrm{id}_I"'] & J \arrow[d, "t"'] \arrow[dd, bend left=60, "\mathrm{id}_J"] \\
Z \arrow[r, "h"'] \arrow[d, "\xi"] \arrow[ur, bend left] & X \arrow[r, "f"] \arrow[d, "\varphi"] \arrow[ur, bend left, "g"] \arrow[dr, phantom, "\lrcorner"] & Y \arrow[d, "\psi"] \\
K \arrow[r, "v"] & I \arrow[r, "u"] & J
\end{tikzcd}$$

It follows that one choice of "?" filler in (1.2.3.3) is $? = (v, h)$, and its uniqueness follows from the uniqueness of h since $I \xleftarrow{\varphi} X \xrightarrow{f} Y$ is a pullback. Thus, (1.2.3.2) is Cartesian, and hence $\mathbb{B}^{\to}_{\bullet} \to \mathbb{B}$ is a fibration. $\square$

Solution, (ii). We define functors

$$F : \mathrm{Fam}(\mathbf{Sets}_\bullet) \longrightarrow \mathbf{Sets}^{\to}_{\bullet}, \qquad G : \mathbf{Sets}^{\to}_{\bullet} \longrightarrow \mathrm{Fam}(\mathbf{Sets}_\bullet)$$

on objects by

$$F\left((X_i, x_i)_{i\in I}\right) = \left\langle \begin{pmatrix} \sum_{i\in I} X_i \\ \downarrow_{\mathrm{fst}} \\ I \end{pmatrix}, i \longmapsto (i, x_i) \right\rangle$$

and

$$G\left(\left\langle \begin{pmatrix} X \\ \downarrow\varphi \\ I \end{pmatrix}, s \right\rangle\right) = \left(\varphi^{-1}(\{i\}), s(i)\right)_{i\in I}$$

and on morphisms by

$$F\left((X_i, x_i)_{i\in I} \xrightarrow{(u,f)} (Y_j, y_j)_{j\in J}\right) = \left(u, (i,x) \longmapsto (u(i), f_i(x))\right)$$

and

$$G\left(\left\langle \begin{pmatrix} X \\ \downarrow\varphi \\ I \end{pmatrix}, s \right\rangle \xrightarrow{(u,f)} \left\langle \begin{pmatrix} Y \\ \downarrow\psi \\ J \end{pmatrix}, t \right\rangle\right) = \left(u, \left(f|_{\varphi^{-1}(\{i\})}\right)_{i\in I}\right).$$

Then we have

$$G\left(F\left((X_i, x_i)_{i\in I}\right)\right) = \left(\{i\} \times X_i, x_i\right)_{i\in I} \cong (X_i, x_i)_{i\in I}$$

in $\mathrm{Fam}(\mathbf{Sets}_\bullet)$ and

$$F\left(G\left(\left\langle \begin{pmatrix} X \\ \downarrow\varphi \\ I \end{pmatrix}, s \right\rangle\right)\right) = \left\langle \begin{pmatrix} \sum_{i\in I} \varphi^{-1}(\{i\}) \\ \downarrow\mathrm{fst} \\ I \end{pmatrix}, i \longmapsto (i, s(i)) \right\rangle$$
$$\cong \left\langle \begin{pmatrix} X \\ \downarrow\varphi \\ I \end{pmatrix}, s \right\rangle$$

in $\mathbf{Sets}_\bullet^{\to}$. We omit the proofs of the naturality of these isomorphisms, and conclude that $G \circ F \cong \mathrm{id}_{\mathrm{Fam}(\mathbf{Sets}_\bullet)}$ and $F \circ G \cong \mathrm{id}_{\mathbf{Sets}_\bullet^{\to}}$, and that, in particular, there is an equivalence of categories $F : \mathrm{Fam}(\mathbf{Sets}_\bullet) \xrightarrow{\simeq} \mathbf{Sets}_\bullet^{\to}$ □

1.2.4. Check that for a PER R one has $R \subseteq |R|^2$.

Solution. Let R be a partial equivalence relation (on an arbitrary set, not necessarily just $\mathbb{N}$), and let $(x, y) \in R$. Since R is symmetric, we also have $(y, x) \in R$, and since R is transitive, it follows that $(x, x) \in R$ and $(y, y) \in R$. Thus, $x \in |R|$ and $y \in |R|$, so $(x, y) \in |R|^2$. Therefore $R \subseteq |R|^2$. □

1.2.5. Let I be a set. A **partition** of I is a collection $Q \subseteq P(I)$ of subsets of I satisfying (1) every set in Q is non-empty (2) if $a, b \in Q$ and $a \cap b \neq \emptyset$, then $a = b$ (3) $\bigcup Q = I$. A **partial partition** of I is a subset $Q \subseteq P(I)$ satisfying (1) and (2) but not necessarily (3). Show that

(i) there is a bijective correspondence between partitions and equivalence relations and between partial partitions and partial equivalence relations (on I);

(ii) there is a bijective correspondence between partial equivalence relations on I and equivalence relations on subsets of I.

Solution, (i). For a set I, let $\mathrm{PPart}(I)$ and $\mathrm{PER}(I)$ denote the sets of partial partitions on I and partial equivalence relations on I, respectively. Within these, let $\mathrm{Part}(I) \subseteq \mathrm{PPart}(I)$ and $\mathrm{ER}(I) \subseteq \mathrm{PER}(I)$ denote the sets of (total) partitions on I and (total) equivalence relations on I, respectively. Define

$$f : \mathrm{PPart}(I) \longrightarrow \mathrm{PER}(I), \qquad g : \mathrm{PER}(I) \longrightarrow \mathrm{PPart}(I)$$

by

$$\begin{aligned} f(Q) &= \Big\{(x, y) \in I^2 : \text{there exists } a \in Q \text{ such that } x, y \in a\Big\}, \\ g(R) &= \Big\{\text{equivalence classes of } R\Big\}. \end{aligned}$$

A routine verification shows that f and g are mutual inverses, and furthermore they restrict to mutually inverse bijections between $\mathrm{Part}(I)$ and $\mathrm{ER}(I)$. $\square$

Solution, (ii). Note that for distinct subsets J and K of I, the sets $\mathrm{ER}(J)$ and $\mathrm{ER}(K)$ are disjoint, so we have

$$\mathrm{PER}(I) = \bigcup_{J \subseteq I} \mathrm{ER}(J) \cong \coprod_{J \subseteq I} \mathrm{ER}(J). \qquad \square$$

1.2.6. Notice that for $R \in \mathbf{PER}$, the "global sections" or "global elements" homset $\mathbf{PER}(1, R)$ is isomorphic to the quotient $\mathbb{N}/R$. And also that all homsets in $\mathbf{PER}$ and in ω-$\mathbf{Sets}$ are countable.

Proof. First, fix $R \in \mathbf{PER}$. For a terminal object 1 in $\mathbf{PER}$, choose $1 = \{(0,0)\}$, and denote $\mathbb{N}/1 = \{[0]_1\} = *$. Define $\varphi : \mathbf{PER}(1, R) \to \mathbb{N}/R$ by

$$\varphi\left(* \xrightarrow{f} \mathbb{N}/R\right) = f([0]_1).$$

Then φ is injective (since $[0]_1$ is the only element of $*$). Moreover, take an equivalence class $[n]_R \in \mathbb{N}/R$ for some $n \in |R|$. Let $e \in \mathbb{N}$ be the code of the constant total recursive function $\mathbb{N} \to \mathbb{N}$, $_ \mapsto n$. Then define $f : * \to \mathbb{N}/R$ by $f([0]_1) = [n]_R$, and note that f is tracked by e, so $f \in \mathbf{PER}(1, R)$. Then we have $\varphi(f) = [n]_R$, so φ is surjective. Thus, φ is an isomorphism (in $\mathbf{Sets}$) between $\mathbf{PER}(1, R)$ and $\mathbb{N}/R$.

Lastly, to see that all homsets in $\mathbf{PER}$ and ω-$\mathbf{Sets}$ are countable, note that all the morphisms in these two categories are tracked by partial recursive functions, and there are only countably many partially recursive functions. □

1.2.7.

(i) Prove that for each ω-set (I, E) the slice category ω-$\mathbf{Sets}/(I, E)$ is Cartesian closed, *i.e.* that ω-$\mathbf{Sets}$ is a locally Cartesian closed category (LCCC).

(ii) Show that also $\mathbf{PER}$ is an LCCC.

1.2.8. Show that a map $\nabla X \to (\mathbb{N}/R, \in)$ in ω-$\mathbf{Sets}$ is constant (where X is a set and R is a PER).

Solution. Let X be a set, R a PER, and $f : \nabla X \to (\mathbb{N}/R, \in)$ a morphism in ω-$\mathbf{Sets}$. Choose a code $e \in \mathbb{N}$ that tracks f. Then for all $x \in X$ and $n \in \mathbb{N}$, $e \cdot n$ is defined and $e \cdot n \in f(x) \in \mathbb{N}/R$. Since R is a PER, this implies that $f(x) = f(y)$ for all $x, y \in X$, so f is constant. □

1.6 Fibrations of signatures

1.6.1. Write down a (single-typed) signature for groups and also a (many-typed) signature for vector spaces.

Solution, groups. A possible single-typed signature for groups is given by a basic type G and function symbols

$$\begin{aligned} \mathsf{op} &: \mathsf{G}, \mathsf{G} \longrightarrow \mathsf{G}, \\ \mathsf{id} &: () \longrightarrow \mathsf{G}, \\ \mathsf{inv} &: \mathsf{G} \longrightarrow \mathsf{G}, \end{aligned}$$

representing the group operation, unit element, and inversion, respectively. □

Solution, vector spaces. It is not clear how to represent multiplicative inversion in a field—which is a partial function—so we will cheat and model (left) modules over a ring instead. A possible multi-typed signature for modules is given by two basic types R (the ring) and M (the abelian group on which the ring acts) and function symbols taken from the following collections.

(a) Function symbols representing the intrinsic ring attributes

$$\begin{aligned} \mathsf{add} &: \mathsf{R}, \mathsf{R} \longrightarrow \mathsf{R}, \\ \mathsf{multiply} &: \mathsf{R}, \mathsf{R} \longrightarrow \mathsf{R}, \\ 0 &: () \longrightarrow \mathsf{R}, \\ 1 &: () \longrightarrow \mathsf{R}, \\ \mathsf{negate} &: \mathsf{R} \longrightarrow \mathsf{R} \end{aligned}$$

(respectively, addition, multiplication, zero, one, and negation).

(b) Function symbols representing the intrinsic abelian group attributes

$$\begin{aligned} \mathsf{add} &: \mathsf{M}, \mathsf{M} \longrightarrow \mathsf{M}, \\ 0 &: () \longrightarrow \mathsf{M}, \\ \mathsf{negate} &: \mathsf{M} \longrightarrow \mathsf{M} \end{aligned}$$

(respectively, addition, zero, and negation).

(c) A single function symbol

$$\mathsf{multiply} : \mathsf{R}, \mathsf{M} \longrightarrow \mathsf{M}$$

representing the action of the ring on the abelian group. □

1.6.2. A many-typed signature is called **finite** if it has only finitely many types and function symbols. Define the subcategory of $\mathbf{FinSign} \hookrightarrow \mathbf{Sign}$ consisting of finite signatures by change-of-base.

Solution. Recall that **Sign** can be defined via the change-of-base situation in the bottom right square of the following commutative diagram in **Cat**.

$$\begin{tikzcd}
 & & \mathrm{FinFam}(\mathbf{Sets}) \arrow[d, hook] \\
 & \mathbf{Sign} \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & \mathrm{Fam}(\mathbf{Sets}) \arrow[d] \\
\mathbf{FinSets} \arrow[r, hook] & \mathbf{Sets} \arrow[r, "T \mapsto T^* \times T"'] & \mathbf{Sets}
\end{tikzcd}$$

This situation can be completed by pullbacks in **Cat** as follows (noting that pullbacks of monic morphisms are monic; cf., e.g., [2, Proposition 2.5.3]).

$$\begin{tikzcd}
\mathbf{FinSign} \arrow[r, hook] \arrow[d, hook] \arrow[dr, phantom, "\lrcorner", very near start] & _ \arrow[r] \arrow[d, hook] \arrow[dr, phantom, "\lrcorner", very near start] & \mathrm{FinFam}(\mathbf{Sets}) \arrow[d, hook] \\
_ \arrow[r, hook] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & \mathbf{Sign} \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & \mathrm{Fam}(\mathbf{Sets}) \arrow[d] \\
\mathbf{FinSets} \arrow[r, hook] & \mathbf{Sets} \arrow[r, "T \mapsto T^* \times T"'] & \mathbf{Sets}
\end{tikzcd}$$

This presentation recovers **FinSign** as a subcategory of **Sign**, and the outer square is a pullback by Exercise 1.1.5, which simplifies to the following change-of-base situation.

$$\begin{tikzcd}
\mathbf{FinSign} \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & \mathrm{FinFam}(\mathbf{Sets}) \arrow[d] \\
\mathbf{FinSets} \arrow[r, "T \mapsto T^* \times T"'] & \mathbf{Sets}
\end{tikzcd}$$

$\square$

1.6.3. Many-typed signatures are sometimes defined (like in [343] or in [282, 2.2.1]) as objects of the category $\mathbf{Sign}'$ which arises in the following change-of-base situation.

$$\begin{tikzcd} \mathbf{Sign}' \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & \mathbf{Sets}^{\to} \arrow[d, "\mathrm{cod}"] \\ \mathbf{Sets} \arrow[r, "T \mapsto T^* \times T"'] & \mathbf{Sets} \end{tikzcd}$$

(i) Describe the category $\mathbf{Sign}'$ in elementary terms.

(ii) Show that the categories $\mathbf{Sign}$ and $\mathbf{Sign}'$ are equivalent.

(iii) One often prefers $\mathbf{Sign}$ to $\mathbf{Sign}'$ because signatures in $\mathbf{Sign}$ allow **overloading** of function symbols: for example the use of $+$ both for addition of integers and for addition of reals. Explain.

[Another advantage of $\mathbf{Sign}$ is that $\begin{array}{c}\mathbf{Sign}\\ \downarrow \\ \mathbf{Sets}\end{array}$ is a *split* fibration.]

Solution, (i). The category $\mathbf{Sign}'$ has

objects Pairs $\Sigma = (T, \mathcal{F})$, where T is an ordinary set of **basic types** and

$$\mathcal{F} : X \longrightarrow T^* \times T,$$

is a "display-indexed" family of function symbols with total space X. If $f \in X$ and $\mathcal{F}(f) = \langle\langle \sigma_0, \ldots, \sigma_{n-1} \rangle, \sigma_n \rangle \in T^* \times T$, then f is a function symbol with n inputs of types $\sigma_0, \ldots, \sigma_{n-1}$ and an output of type σ_n; this is written

$$f : \sigma_0,\ \ldots,\ \sigma_{n-1} \longrightarrow \sigma_n.$$

The fibre $\mathcal{F}^{-1}\left(\langle\langle \sigma_0, \ldots, \sigma_{n-1} \rangle, \sigma_n \rangle\right)$ over the index $\langle\langle \sigma_0, \ldots, \sigma_{n-1} \rangle, \sigma_n \rangle$ is the set of all such function symbols.

morphisms Given signatures $\Sigma = (T, \mathcal{F})$ and $\Sigma' = (S, \mathcal{G})$ in $\mathbf{Sign}'$, where

$$\mathcal{F} : X \longrightarrow T^* \times T, \qquad \mathcal{G} : Y \longrightarrow S^* \times S,$$

a morphism from Σ to Σ' is a pair (u, φ) where $u : T \to S$ and $\varphi : X \to Y$ are functions such that the following diagram commutes

$$\begin{tikzcd} X \arrow[r, "\varphi"] \arrow[d, "\mathcal{F}"'] & Y \arrow[d, "\mathcal{G}"] \\ T^* \times T \arrow[r, "u^* \times u"] & S^* \times S \end{tikzcd}$$

That is, if $f \in X$ with $f : \sigma_1, \ldots, \sigma_{n-1} \to \sigma_n$ (in Σ), then

$$\varphi(f) : u(\sigma_1), \ldots, u(\sigma_{n-1}) \longrightarrow u(\sigma_n) \qquad \text{(in } \Sigma'\text{)} \qquad \square$$

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